Newman-Penrose Formalism Calculations

Stephani et al., Equation (15.38_k1)

Metric

$$ds^2 = -\mathrm{d}t \otimes \mathrm{d}t + \left(-\frac{\left(\frac{\partial Y}{\partial r}\right)^2}{\epsilon f^2 - k}\right) \mathrm{d}r \otimes \mathrm{d}r + Y^2 \mathrm{d}\theta \otimes \mathrm{d}\theta + Y^2 \sin(\theta)^2 \mathrm{d}\phi \otimes \mathrm{d}\phi$$

Coordinates

$$(x^\mu) = (t, r, \theta, \phi)$$

Metric Functions

$$f(\theta) = f$$

$$Y(r, t) = Y$$

Null Tetrad

Covariant components

$$l = \left(\sqrt{\frac{1}{2}}\right) dt + \left(\sqrt{\frac{1}{2}} \sqrt{-\frac{1}{\epsilon f^2 - k}} \frac{\partial}{\partial r} Y\right) dr$$

$$n = \left(\sqrt{\frac{1}{2}}\right) dt + \left(-\sqrt{\frac{1}{2}} \sqrt{-\frac{1}{\epsilon f^2 - k}} \frac{\partial}{\partial r} Y\right) dr$$

$$m = \left(\frac{1}{2} \sqrt{2} Y\right) d\theta + \left(\frac{1}{2} i \sqrt{2} Y \sin(\theta)\right) d\phi$$

$$\overline{m} = \left(\frac{1}{2}\sqrt{2}Y\right) d\theta + \left(-\frac{1}{2}i\sqrt{2}Y\sin(\theta)\right) d\phi$$

Contravariant components

$$l = \left(-\frac{1}{2}\sqrt{2}\right) dt + \left(\frac{\sqrt{2}\sqrt{-\epsilon f^2 + k}}{2\frac{\partial}{\partial r}Y}\right) dr$$

$$n = \left(-\frac{1}{2}\sqrt{2}\right) dt + \left(\frac{\sqrt{2}\epsilon f^2}{2\sqrt{-\epsilon f^2 + k}\frac{\partial}{\partial r}Y} - \frac{\sqrt{2}k}{2\sqrt{-\epsilon f^2 + k}\frac{\partial}{\partial r}Y}\right) dr$$

$$m = \left(\frac{\sqrt{2}}{2Y}\right) d\theta + \left(\frac{i\sqrt{2}}{2Y\sin(\theta)}\right) d\phi$$

$$\overline{m} = \left(\frac{\sqrt{2}}{2Y}\right) d\theta + \left(-\frac{i\sqrt{2}}{2Y\sin(\theta)}\right) d\phi$$

Directional Derivatives

$$D = -\frac{1}{2}\sqrt{2}\,\partial_t + \frac{\sqrt{2}\sqrt{-\epsilon f^2 + k}}{2\frac{\partial}{\partial r}Y}\,\partial_r$$

$$\Delta = -\frac{1}{2}\sqrt{2}\,\partial_t + \frac{\sqrt{2}\epsilon f^2}{2\sqrt{-\epsilon f^2 + k}\frac{\partial}{\partial r}Y} - \frac{\sqrt{2}k}{2\sqrt{-\epsilon f^2 + k}\frac{\partial}{\partial r}Y}\,\partial_r$$

$$\delta = \frac{\sqrt{2}}{2Y}\,\partial_\theta + \frac{i\sqrt{2}}{2Y\sin(\theta)}\,\partial_\phi$$

$$\overline{\delta} = \frac{\sqrt{2}}{2Y}\,\partial_\theta - \frac{i\sqrt{2}}{2Y\sin(\theta)}\,\partial_\phi$$

Spin Coefficients

$$\rho = \frac{\sqrt{2}\epsilon f\left(\theta\right)^{2}}{2\sqrt{-\epsilon f\left(\theta\right)^{2}+kY}} + \frac{\sqrt{2}\frac{\partial}{\partial t}Y}{2\,Y} - \frac{\sqrt{2}k}{2\sqrt{-\epsilon f\left(\theta\right)^{2}+kY}}$$

$$\sigma = 0$$

$$\kappa = -\frac{\epsilon f\left(\theta\right)\frac{\partial}{\partial\theta}f\left(\theta\right)}{2\left(\sqrt{2}\epsilon Y f\left(\theta\right)^{2}-\sqrt{2}kY\right)}$$

$$\mu = \frac{\sqrt{2}\epsilon f\left(\theta\right)^{2}}{2\sqrt{-\epsilon f\left(\theta\right)^{2}+kY}} - \frac{\sqrt{2}\frac{\partial}{\partial t}Y}{2\,Y} - \frac{\sqrt{2}k}{2\sqrt{-\epsilon f\left(\theta\right)^{2}+kY}}$$

$$\lambda = 0$$

$$\nu = \frac{\epsilon f\left(\theta\right)\frac{\partial}{\partial\theta}f\left(\theta\right)}{2\left(\sqrt{2}\epsilon Y f\left(\theta\right)^{2}-\sqrt{2}kY\right)}$$

$$\alpha = -\frac{\sqrt{2}\cos\left(\theta\right)}{4\,Y\sin\left(\theta\right)}$$

$$\beta = \frac{\sqrt{2}\cos\left(\theta\right)}{4\,Y\sin\left(\theta\right)}$$

$$\tau = \frac{\epsilon f\left(\theta\right)\frac{\partial}{\partial\theta}f\left(\theta\right)}{2\left(\sqrt{2}\epsilon Y f\left(\theta\right)^{2}-\sqrt{2}kY\right)}$$

$$\pi = -\frac{\epsilon f(\theta) \frac{\partial}{\partial\theta} f(\theta)}{2\left(\sqrt{2}\epsilon Y f(\theta)^2 - \sqrt{2}kY\right)}$$

$$\epsilon = -\frac{\sqrt{2}\frac{\partial^2}{\partial r\partial t}Y}{4\frac{\partial}{\partial r}Y}$$

$$\gamma = \frac{\sqrt{2}\frac{\partial^2}{\partial r\partial t}Y}{4\frac{\partial}{\partial r}Y}$$

Ricci Tensor Components

$$\begin{aligned}
\Phi_{00} = {} & \frac{\epsilon^2 Y f(\theta)^4 \frac{\partial^2}{\partial r\partial t}Y \frac{\partial}{\partial t}Y}{2\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r}Y - 2\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r}Y + k^2 Y^2 \frac{\partial}{\partial r}Y\right)} - \frac{\epsilon^2 Y f(\theta)^4 \frac{\partial}{\partial r}Y \frac{\partial^2}{(\partial t)^2}Y}{2\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r}Y - 2\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r}Y + k^2 Y^2 \frac{\partial}{\partial r}Y\right)} - \frac{\epsilon k Y f(\theta)^2 \frac{\partial^2}{\partial r\partial t}Y \frac{\partial}{\partial t}Y}{\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r}Y - 2\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r}Y + k^2 Y^2 \frac{\partial}{\partial r}Y} \\
& + \frac{\epsilon k Y f(\theta)^2 \frac{\partial}{\partial r}Y \frac{\partial^2}{(\partial t)^2}Y}{\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r}Y - 2\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r}Y + k^2 Y^2 \frac{\partial}{\partial r}Y} + \frac{\epsilon^2 \cos(\theta) f(\theta)^3 \frac{\partial}{\partial r}Y \frac{\partial}{\partial\theta} f(\theta)}{4\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r}Y - 2\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r}Y + k^2 Y^2 \frac{\partial}{\partial r}Y\right)\sin(\theta)} \\
& - \frac{\epsilon^2 f(\theta)^2 \frac{\partial}{\partial r}Y \frac{\partial}{\partial\theta} f(\theta)^2}{2\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r}Y - 2\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r}Y + k^2 Y^2 \frac{\partial}{\partial r}Y\right)} + \frac{\epsilon^2 f(\theta)^3 \frac{\partial}{\partial r}Y \frac{\partial^2}{(\partial\theta)^2} f(\theta)}{4\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r}Y - 2\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r}Y + k^2 Y^2 \frac{\partial}{\partial r}Y\right)} + \frac{k^2 Y \frac{\partial^2}{\partial r\partial t}Y \frac{\partial}{\partial t}Y}{2\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r}Y - 2\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r}Y + k^2 Y^2 \frac{\partial}{\partial r}Y\right)} \\
& - \frac{k^2 Y \frac{\partial}{\partial r}Y \frac{\partial^2}{(\partial t)^2}Y}{2\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r}Y - 2\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r}Y + k^2 Y^2 \frac{\partial}{\partial r}Y\right)} - \frac{\epsilon k \cos(\theta) f(\theta) \frac{\partial}{\partial r}Y \frac{\partial}{\partial\theta} f(\theta)}{4\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r}Y - 2\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r}Y + k^2 Y^2 \frac{\partial}{\partial r}Y\right)\sin(\theta)} \\
& - \frac{\epsilon k \frac{\partial}{\partial r}Y \frac{\partial}{\partial\theta} f(\theta)^2}{4\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r}Y - 2\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r}Y + k^2 Y^2 \frac{\partial}{\partial r}Y\right)} - \frac{\epsilon k f(\theta) \frac{\partial}{\partial r}Y \frac{\partial^2}{(\partial\theta)^2} f(\theta)}{4\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r}Y - 2\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r}Y + k^2 Y^2 \frac{\partial}{\partial r}Y\right)}
\end{aligned}$$

$$\Phi_{01} = -\frac{\epsilon Y f(\theta) \frac{\partial^2}{\partial r\partial t}Y \frac{\partial}{\partial\theta} f(\theta)}{4\left(\epsilon Y^2 f(\theta)^2 \frac{\partial}{\partial r}Y - kY^2 \frac{\partial}{\partial r}Y\right)} + \frac{\epsilon f(\theta) \frac{\partial}{\partial r}Y \frac{\partial}{\partial t}Y \frac{\partial}{\partial\theta} f(\theta)}{4\left(\epsilon Y^2 f(\theta)^2 \frac{\partial}{\partial r}Y - kY^2 \frac{\partial}{\partial r}Y\right)} - \frac{\sqrt{-\epsilon f(\theta)^2 + k}\epsilon f(\theta) \frac{\partial}{\partial r}Y \frac{\partial}{\partial\theta} f(\theta)}{4\left(\epsilon Y^2 f(\theta)^2 \frac{\partial}{\partial r}Y - kY^2 \frac{\partial}{\partial r}Y\right)}$$

$$\Phi_{02} = -\frac{\epsilon^2 \cos(\theta) f(\theta)^3 \frac{\partial}{\partial\theta} f(\theta)}{4\left(\epsilon^2 Y^2 f(\theta)^4 - 2\,\epsilon k Y^2 f(\theta)^2 + k^2 Y^2\right)\sin(\theta)} - \frac{\epsilon^2 f(\theta)^2 \frac{\partial}{\partial\theta} f(\theta)^2}{2\left(\epsilon^2 Y^2 f(\theta)^4 - 2\,\epsilon k Y^2 f(\theta)^2 + k^2 Y^2\right)} + \frac{\epsilon^2 f(\theta)^3 \frac{\partial^2}{(\partial\theta)^2} f(\theta)}{4\left(\epsilon^2 Y^2 f(\theta)^4 - 2\,\epsilon k Y^2 f(\theta)^2 + k^2 Y^2\right)}$$
$$+ \frac{\epsilon k \cos(\theta) f(\theta) \frac{\partial}{\partial\theta} f(\theta)}{4\left(\epsilon^2 Y^2 f(\theta)^4 - 2\,\epsilon k Y^2 f(\theta)^2 + k^2 Y^2\right)\sin(\theta)} - \frac{\epsilon k \frac{\partial}{\partial\theta} f(\theta)^2}{4\left(\epsilon^2 Y^2 f(\theta)^4 - 2\,\epsilon k Y^2 f(\theta)^2 + k^2 Y^2\right)} - \frac{\epsilon k f(\theta) \frac{\partial^2}{(\partial\theta)^2} f(\theta)}{4\left(\epsilon^2 Y^2 f(\theta)^4 - 2\,\epsilon k Y^2 f(\theta)^2 + k^2 Y^2\right)}$$

$$\Phi_{10} = -\frac{\epsilon Y f(\theta) \frac{\partial^2}{\partial r \partial t} Y \frac{\partial}{\partial\theta} f(\theta)}{4\left(\epsilon Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y - k Y^2 \frac{\partial}{\partial r} Y\right)} + \frac{\epsilon f(\theta) \frac{\partial}{\partial r} Y \frac{\partial}{\partial t} Y \frac{\partial}{\partial\theta} f(\theta)}{4\left(\epsilon Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y - k Y^2 \frac{\partial}{\partial r} Y\right)} - \frac{\sqrt{-\epsilon f(\theta)^2 + k}\,\epsilon f(\theta) \frac{\partial}{\partial r} Y \frac{\partial}{\partial\theta} f(\theta)}{4\left(\epsilon Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y - k Y^2 \frac{\partial}{\partial r} Y\right)}$$

$$\Phi_{11} = \frac{\epsilon f(\theta)^2}{4\,Y^2} - \frac{\frac{\partial^3}{\partial r (\partial t)^2} Y}{4 \frac{\partial}{\partial r} Y} + \frac{\frac{\partial}{\partial t} Y^2}{4\,Y^2} - \frac{k}{4\,Y^2} + \frac{1}{4\,Y^2}$$

$$\Phi_{12} = -\frac{\epsilon^2 f(\theta)^3 \frac{\partial}{\partial r} Y \frac{\partial}{\partial\theta} f(\theta)}{4\left(\epsilon Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y - k Y^2 \frac{\partial}{\partial r} Y\right)\sqrt{-\epsilon f(\theta)^2 + k}} - \frac{\epsilon Y f(\theta) \frac{\partial^2}{\partial r \partial t} Y \frac{\partial}{\partial\theta} f(\theta)}{4\left(\epsilon Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y - k Y^2 \frac{\partial}{\partial r} Y\right)} + \frac{\epsilon f(\theta) \frac{\partial}{\partial r} Y \frac{\partial}{\partial t} Y \frac{\partial}{\partial\theta} f(\theta)}{4\left(\epsilon Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y - k Y^2 \frac{\partial}{\partial r} Y\right)} + \frac{\epsilon k f(\theta) \frac{\partial}{\partial r} Y \frac{\partial}{\partial\theta} f(\theta)}{4\left(\epsilon Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y - k Y^2 \frac{\partial}{\partial r} Y\right)\sqrt{-\epsilon f(\theta)^2 + k}}$$

$$\Phi_{20} = -\frac{\epsilon^2 \cos(\theta) f(\theta)^3 \frac{\partial}{\partial\theta} f(\theta)}{4\left(\epsilon^2 Y^2 f(\theta)^4 - 2\,\epsilon k Y^2 f(\theta)^2 + k^2 Y^2\right)\sin(\theta)} - \frac{\epsilon^2 f(\theta)^2 \frac{\partial}{\partial\theta} f(\theta)^2}{2\left(\epsilon^2 Y^2 f(\theta)^4 - 2\,\epsilon k Y^2 f(\theta)^2 + k^2 Y^2\right)} + \frac{\epsilon^2 f(\theta)^3 \frac{\partial^2}{(\partial\theta)^2} f(\theta)}{4\left(\epsilon^2 Y^2 f(\theta)^4 - 2\,\epsilon k Y^2 f(\theta)^2 + k^2 Y^2\right)}$$
$$+ \frac{\epsilon k \cos(\theta) f(\theta) \frac{\partial}{\partial\theta} f(\theta)}{4\left(\epsilon^2 Y^2 f(\theta)^4 - 2\,\epsilon k Y^2 f(\theta)^2 + k^2 Y^2\right)\sin(\theta)} - \frac{\epsilon k \frac{\partial}{\partial\theta} f(\theta)^2}{4\left(\epsilon^2 Y^2 f(\theta)^4 - 2\,\epsilon k Y^2 f(\theta)^2 + k^2 Y^2\right)} - \frac{\epsilon k f(\theta) \frac{\partial^2}{(\partial\theta)^2} f(\theta)}{4\left(\epsilon^2 Y^2 f(\theta)^4 - 2\,\epsilon k Y^2 f(\theta)^2 + k^2 Y^2\right)}$$

$$\Phi_{21} = -\frac{\epsilon^2 f(\theta)^3 \frac{\partial}{\partial r} Y \frac{\partial}{\partial\theta} f(\theta)}{4\left(\epsilon Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y - k Y^2 \frac{\partial}{\partial r} Y\right)\sqrt{-\epsilon f(\theta)^2 + k}} - \frac{\epsilon Y f(\theta) \frac{\partial^2}{\partial r \partial t} Y \frac{\partial}{\partial\theta} f(\theta)}{4\left(\epsilon Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y - k Y^2 \frac{\partial}{\partial r} Y\right)} + \frac{\epsilon f(\theta) \frac{\partial}{\partial r} Y \frac{\partial}{\partial t} Y \frac{\partial}{\partial\theta} f(\theta)}{4\left(\epsilon Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y - k Y^2 \frac{\partial}{\partial r} Y\right)} + \frac{\epsilon k f(\theta) \frac{\partial}{\partial r} Y \frac{\partial}{\partial\theta} f(\theta)}{4\left(\epsilon Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y - k Y^2 \frac{\partial}{\partial r} Y\right)\sqrt{-\epsilon f(\theta)^2 + k}}$$

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\begin{aligned}
\Phi_{22} = {} & \frac{\epsilon^2 Y f(\theta)^4 \frac{\partial^2}{\partial r \partial t} Y \frac{\partial}{\partial t} Y}{2\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\,\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} - \frac{\epsilon^2 Y f(\theta)^4 \frac{\partial}{\partial r} Y \frac{\partial^2}{(\partial t)^2} Y}{2\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\,\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} - \frac{\epsilon k Y f(\theta)^2 \frac{\partial^2}{\partial r \partial t} Y \frac{\partial}{\partial t} Y}{\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\,\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y} \\
& + \frac{\epsilon k Y f(\theta)^2 \frac{\partial}{\partial r} Y \frac{\partial^2}{(\partial t)^2} Y}{\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\,\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y} + \frac{\epsilon^2 \cos(\theta) f(\theta)^3 \frac{\partial}{\partial r} Y \frac{\partial}{\partial \theta} f(\theta)}{4\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\,\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)\sin(\theta)} \\
& - \frac{\epsilon^2 f(\theta)^2 \frac{\partial}{\partial r} Y \frac{\partial}{\partial \theta} f(\theta)^2}{2\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\,\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} + \frac{\epsilon^2 f(\theta)^3 \frac{\partial}{\partial r} Y \frac{\partial^2}{(\partial \theta)^2} f(\theta)}{4\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\,\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} + \frac{k^2 Y \frac{\partial^2}{\partial r \partial t} Y \frac{\partial}{\partial t} Y}{2\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\,\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} \\
& - \frac{k^2 Y \frac{\partial}{\partial r} Y \frac{\partial^2}{(\partial t)^2} Y}{2\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\,\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} - \frac{\epsilon k \cos(\theta) f(\theta) \frac{\partial}{\partial r} Y \frac{\partial}{\partial \theta} f(\theta)}{4\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\,\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)\sin(\theta)} \\
& - \frac{\epsilon k \frac{\partial}{\partial r} Y \frac{\partial}{\partial \theta} f(\theta)^2}{4\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\,\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} - \frac{\epsilon k f(\theta) \frac{\partial}{\partial r} Y \frac{\partial^2}{(\partial \theta)^2} f(\theta)}{4\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\,\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)}
\end{aligned}
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\begin{aligned}
\Lambda = {} & \frac{\epsilon^3 f(\theta)^6 \frac{\partial}{\partial r} Y}{12\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\,\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} + \frac{\epsilon^2 Y^2 f(\theta)^4 \frac{\partial^3}{\partial r (\partial t)^2} Y}{12\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\,\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} + \frac{\epsilon^2 Y f(\theta)^4 \frac{\partial^2}{\partial r \partial t} Y \frac{\partial}{\partial t} Y}{6\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\,\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} \\
& + \frac{\epsilon^2 f(\theta)^4 \frac{\partial}{\partial r} Y \frac{\partial}{\partial t} Y^2}{12\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\,\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} + \frac{\epsilon^2 Y f(\theta)^4 \frac{\partial}{\partial r} Y \frac{\partial^2}{(\partial t)^2} Y}{6\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\,\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} \\
& - \frac{\epsilon^2 k f(\theta)^4 \frac{\partial}{\partial r} Y}{4\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\,\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} + \frac{\epsilon^2 f(\theta)^4 \frac{\partial}{\partial r} Y}{12\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\,\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} - \frac{\epsilon k Y^2 f(\theta)^2 \frac{\partial^3}{\partial r (\partial t)^2} Y}{6\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\,\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} \\
& - \frac{\epsilon k Y f(\theta)^2 \frac{\partial^2}{\partial r \partial t} Y \frac{\partial}{\partial t} Y}{3\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\,\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} - \frac{\epsilon k f(\theta)^2 \frac{\partial}{\partial r} Y \frac{\partial}{\partial t} Y^2}{6\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\,\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} - \frac{\epsilon k Y f(\theta)^2 \frac{\partial}{\partial r} Y \frac{\partial^2}{(\partial t)^2} Y}{3\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\,\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} \\
& + \frac{\epsilon^2 \cos(\theta) f(\theta)^3 \frac{\partial}{\partial r} Y \frac{\partial}{\partial \theta} f(\theta)}{12\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\,\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)\sin(\theta)} - \frac{\epsilon^2 f(\theta)^2 \frac{\partial}{\partial r} Y \frac{\partial}{\partial \theta} f(\theta)^2}{6\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\,\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} \\
& + \frac{\epsilon^2 f(\theta)^3 \frac{\partial}{\partial r} Y \frac{\partial^2}{(\partial \theta)^2} f(\theta)}{12\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\,\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} + \frac{\epsilon k^2 f(\theta)^2 \frac{\partial}{\partial r} Y}{4\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\,\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} - \frac{\epsilon k f(\theta)^2 \frac{\partial}{\partial r} Y}{6\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\,\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} \\
& + \frac{k^2 Y^2 \frac{\partial^3}{\partial r (\partial t)^2} Y}{12\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\,\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} + \frac{k^2 Y \frac{\partial^2}{\partial r \partial t} Y \frac{\partial}{\partial t} Y}{6\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\,\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} + \frac{k^2 \frac{\partial}{\partial r} Y \frac{\partial}{\partial t} Y^2}{12\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\,\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} \\
& + \frac{k^2 Y \frac{\partial}{\partial r} Y \frac{\partial^2}{(\partial t)^2} Y}{6\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\,\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} - \frac{\epsilon k \cos(\theta) f(\theta) \frac{\partial}{\partial r} Y \frac{\partial}{\partial \theta} f(\theta)}{12\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\,\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)\sin(\theta)} \\
& - \frac{\epsilon k \frac{\partial}{\partial r} Y \frac{\partial}{\partial \theta} f(\theta)^2}{12\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\,\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} - \frac{\epsilon k f(\theta) \frac{\partial}{\partial r} Y \frac{\partial^2}{(\partial \theta)^2} f(\theta)}{12\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\,\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} \\
& - \frac{k^3 \frac{\partial}{\partial r} Y}{12\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\,\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} + \frac{k^2 \frac{\partial}{\partial r} Y}{12\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\,\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)}
\end{aligned}
$$

Weyl Tensor Components

$$
\begin{aligned}
\Psi_0 = {} & -\frac{\epsilon^2 \cos(\theta) f(\theta)^3 \frac{\partial}{\partial \theta} f(\theta)}{4\left(\epsilon^2 Y^2 f(\theta)^4 - 2\,\epsilon k Y^2 f(\theta)^2 + k^2 Y^2\right)\sin(\theta)} - \frac{\epsilon^2 f(\theta)^2 \frac{\partial}{\partial \theta} f(\theta)^2}{2\left(\epsilon^2 Y^2 f(\theta)^4 - 2\,\epsilon k Y^2 f(\theta)^2 + k^2 Y^2\right)} + \frac{\epsilon^2 f(\theta)^3 \frac{\partial^2}{(\partial \theta)^2} f(\theta)}{4\left(\epsilon^2 Y^2 f(\theta)^4 - 2\,\epsilon k Y^2 f(\theta)^2 + k^2 Y^2\right)} \\
& + \frac{\epsilon k \cos(\theta) f(\theta) \frac{\partial}{\partial \theta} f(\theta)}{4\left(\epsilon^2 Y^2 f(\theta)^4 - 2\,\epsilon k Y^2 f(\theta)^2 + k^2 Y^2\right)\sin(\theta)} - \frac{\epsilon k \frac{\partial}{\partial \theta} f(\theta)^2}{4\left(\epsilon^2 Y^2 f(\theta)^4 - 2\,\epsilon k Y^2 f(\theta)^2 + k^2 Y^2\right)} - \frac{\epsilon k f(\theta) \frac{\partial^2}{(\partial \theta)^2} f(\theta)}{4\left(\epsilon^2 Y^2 f(\theta)^4 - 2\,\epsilon k Y^2 f(\theta)^2 + k^2 Y^2\right)}
\end{aligned}
$$

$$\Psi_1 = -\frac{\epsilon Y f(\theta) \frac{\partial^2}{\partial r \partial t} Y \frac{\partial}{\partial \theta} f(\theta)}{4\left(\epsilon Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y - k Y^2 \frac{\partial}{\partial r} Y\right)} + \frac{\epsilon f(\theta) \frac{\partial}{\partial r} Y \frac{\partial}{\partial t} Y \frac{\partial}{\partial \theta} f(\theta)}{4\left(\epsilon Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y - k Y^2 \frac{\partial}{\partial r} Y\right)} + \frac{\sqrt{-\epsilon f(\theta)^2 + k\epsilon} f(\theta) \frac{\partial}{\partial r} Y \frac{\partial}{\partial \theta} f(\theta)}{4\left(\epsilon Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y - k Y^2 \frac{\partial}{\partial r} Y\right)}$$

$$\begin{aligned}
\Psi_2 = &-\frac{\epsilon^3 f(\theta)^6 \frac{\partial}{\partial r} Y}{6\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} - \frac{\epsilon^2 Y^2 f(\theta)^4 \frac{\partial^3}{\partial r (\partial t)^2} Y}{6\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} + \frac{\epsilon^2 Y f(\theta)^4 \frac{\partial^2}{\partial r \partial t} Y \frac{\partial}{\partial t} Y}{6\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} \\
&- \frac{\epsilon^2 f(\theta)^4 \frac{\partial}{\partial r} Y \frac{\partial}{\partial t} Y^2}{6\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} + \frac{\epsilon^2 Y f(\theta)^4 \frac{\partial}{\partial r} Y \frac{\partial^2}{(\partial t)^2} Y}{6\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} + \frac{\epsilon^2 k f(\theta)^4 \frac{\partial}{\partial r} Y}{2\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} \\
&- \frac{\epsilon^2 f(\theta)^4 \frac{\partial}{\partial r} Y}{6\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} + \frac{\epsilon k Y^2 f(\theta)^2 \frac{\partial^3}{\partial r (\partial t)^2} Y}{3\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} - \frac{\epsilon k Y f(\theta)^2 \frac{\partial^2}{\partial r \partial t} Y \frac{\partial}{\partial t} Y}{3\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} \\
&+ \frac{\epsilon k f(\theta)^2 \frac{\partial}{\partial r} Y \frac{\partial}{\partial t} Y^2}{3\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} - \frac{\epsilon k Y f(\theta)^2 \frac{\partial}{\partial r} Y \frac{\partial^2}{(\partial t)^2} Y}{3\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} \\
&+ \frac{\epsilon^2 \cos(\theta) f(\theta)^3 \frac{\partial}{\partial r} Y \frac{\partial}{\partial \theta} f(\theta)}{12\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)\sin(\theta)} - \frac{\epsilon^2 f(\theta)^2 \frac{\partial}{\partial r} Y \frac{\partial}{\partial \theta} f(\theta)^2}{6\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} \\
&+ \frac{\epsilon^2 f(\theta)^3 \frac{\partial}{\partial r} Y \frac{\partial^2}{(\partial \theta)^2} f(\theta)}{12\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} - \frac{\epsilon k^2 f(\theta)^2 \frac{\partial}{\partial r} Y}{2\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} + \frac{\epsilon k f(\theta)^2 \frac{\partial}{\partial r} Y}{3\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} \\
&- \frac{k^2 Y^2 \frac{\partial^3}{\partial r (\partial t)^2} Y}{6\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} + \frac{k^2 Y \frac{\partial^2}{\partial r \partial t} Y \frac{\partial}{\partial t} Y}{6\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} - \frac{k^2 \frac{\partial}{\partial r} Y \frac{\partial}{\partial t} Y^2}{6\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} \\
&+ \frac{k^2 Y \frac{\partial}{\partial r} Y \frac{\partial^2}{(\partial t)^2} Y}{6\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} - \frac{\epsilon k \cos(\theta) f(\theta) \frac{\partial}{\partial r} Y \frac{\partial}{\partial \theta} f(\theta)}{12\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)\sin(\theta)} \\
&- \frac{\epsilon k \frac{\partial}{\partial r} Y \frac{\partial}{\partial \theta} f(\theta)^2}{12\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} - \frac{\epsilon k f(\theta) \frac{\partial}{\partial r} Y \frac{\partial^2}{(\partial \theta)^2} f(\theta)}{12\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} \\
&+ \frac{k^3 \frac{\partial}{\partial r} Y}{6\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)} - \frac{k^2 \frac{\partial}{\partial r} Y}{6\left(\epsilon^2 Y^2 f(\theta)^4 \frac{\partial}{\partial r} Y - 2\epsilon k Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y + k^2 Y^2 \frac{\partial}{\partial r} Y\right)}
\end{aligned}$$

$$\Psi_3 = -\frac{\epsilon Y f(\theta) \frac{\partial^2}{\partial r \partial t} Y \frac{\partial}{\partial \theta} f(\theta)}{4\left(\epsilon Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y - k Y^2 \frac{\partial}{\partial r} Y\right)} + \frac{\epsilon f(\theta) \frac{\partial}{\partial r} Y \frac{\partial}{\partial t} Y \frac{\partial}{\partial \theta} f(\theta)}{4\left(\epsilon Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y - k Y^2 \frac{\partial}{\partial r} Y\right)} - \frac{\sqrt{-\epsilon f(\theta)^2 + k\epsilon} f(\theta) \frac{\partial}{\partial r} Y \frac{\partial}{\partial \theta} f(\theta)}{4\left(\epsilon Y^2 f(\theta)^2 \frac{\partial}{\partial r} Y - k Y^2 \frac{\partial}{\partial r} Y\right)}$$

$$\Psi_4 = -\frac{\epsilon^2 \cos(\theta) f(\theta)^3 \frac{\partial}{\partial\theta} f(\theta)}{4\left(\epsilon^2 Y^2 f(\theta)^4 - 2\,\epsilon k Y^2 f(\theta)^2 + k^2 Y^2\right)\sin(\theta)} - \frac{\epsilon^2 f(\theta)^2 \frac{\partial}{\partial\theta} f(\theta)^2}{2\left(\epsilon^2 Y^2 f(\theta)^4 - 2\,\epsilon k Y^2 f(\theta)^2 + k^2 Y^2\right)} + \frac{\epsilon^2 f(\theta)^3 \frac{\partial^2}{(\partial\theta)^2} f(\theta)}{4\left(\epsilon^2 Y^2 f(\theta)^4 - 2\,\epsilon k Y^2 f(\theta)^2 + k^2 Y^2\right)}$$
$$+ \frac{\epsilon k \cos(\theta) f(\theta) \frac{\partial}{\partial\theta} f(\theta)}{4\left(\epsilon^2 Y^2 f(\theta)^4 - 2\,\epsilon k Y^2 f(\theta)^2 + k^2 Y^2\right)\sin(\theta)} - \frac{\epsilon k \frac{\partial}{\partial\theta} f(\theta)^2}{4\left(\epsilon^2 Y^2 f(\theta)^4 - 2\,\epsilon k Y^2 f(\theta)^2 + k^2 Y^2\right)} - \frac{\epsilon k f(\theta) \frac{\partial^2}{(\partial\theta)^2} f(\theta)}{4\left(\epsilon^2 Y^2 f(\theta)^4 - 2\,\epsilon k Y^2 f(\theta)^2 + k^2 Y^2\right)}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "I"